Research

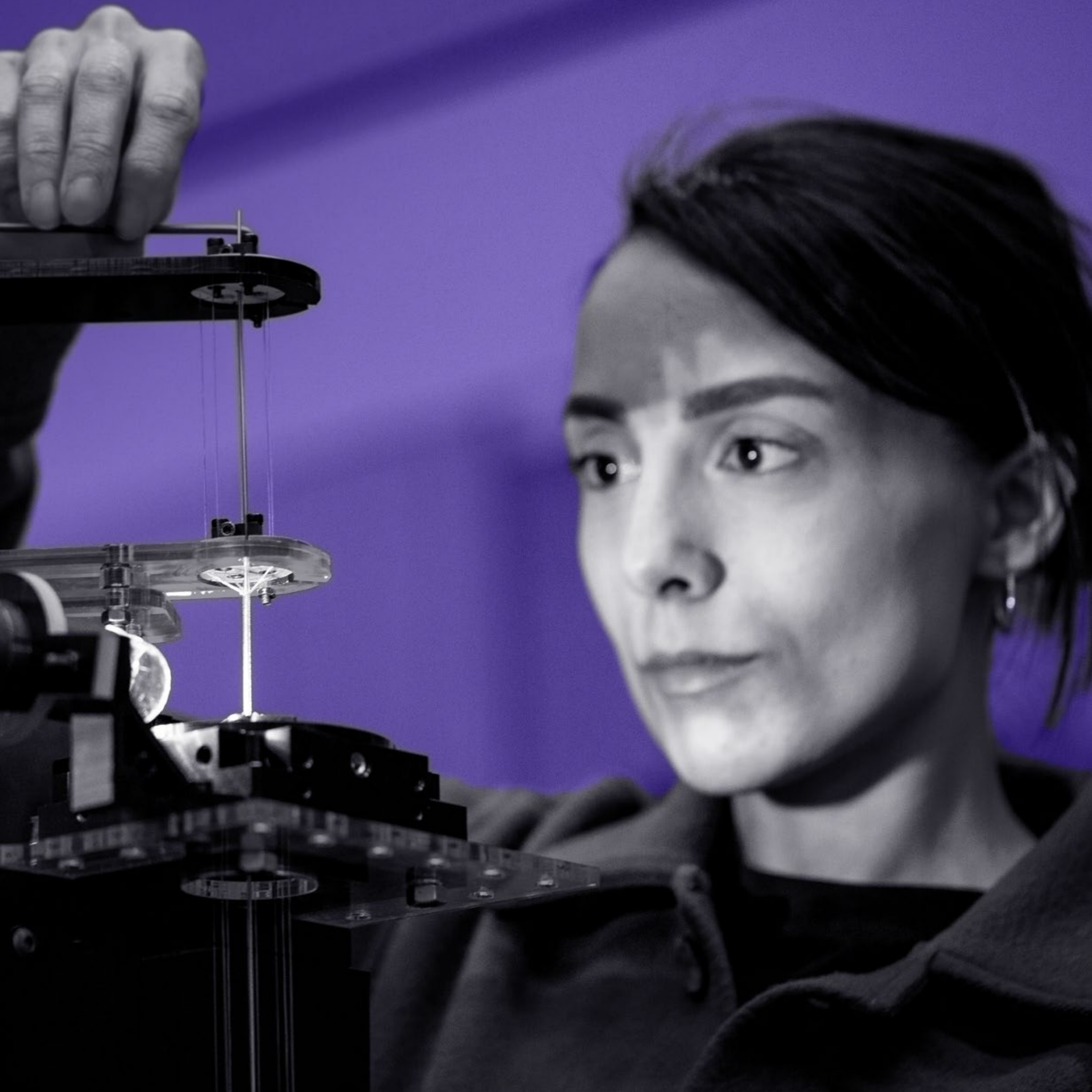
mit
media
lab

Imagine what we can invent

At the MIT Media Lab, research is more than inquiry – it's an act of invention. We are a dynamic, interdisciplinary creative sandbox – grounded in academic excellence – where students, faculty, and collaborators invent transformational futures together.

Our research thrives at the intersection of art, science, engineering, and design. Dozens of research groups, centers, initiatives, and programs pursue hundreds of bold, exploratory projects – from AI to biosensing, and sonic spectaculars to space sustainability.

What unites this diversity is a commitment to rigorous scholarship, hands-on prototyping, and societal relevance. We focus not only on creating and commercializing transformational technologies, but also on understanding and guiding their long-term impact on humanity and the planet – redefining what it means to do impactful academic work in the 21st century.

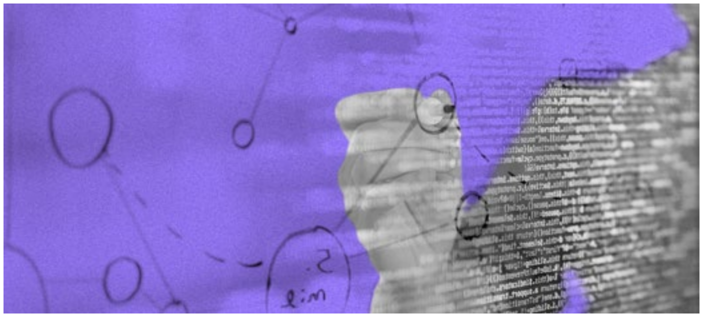


Organized by vision, not disciplines

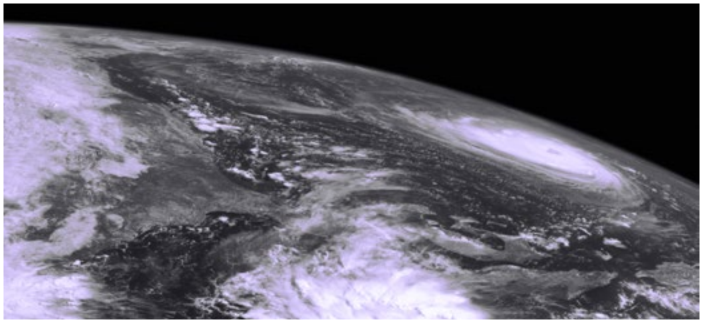
Our research is organized around five visionary research themes that guide the Lab's collective efforts to address the most urgent and inspiring challenges of our time:

Connected Mind + Body: Revolutionizing mental and physical wellbeing through human-centered technologies.

Cultivating Creativity: Empowering people of all ages and backgrounds to unlock their creative potential.



Decentralized Society: Designing open, equitable digital systems for governance, identity, and trust.



Future Worlds: Building sustainable, resilient futures on Earth – and beyond – through transformative innovation.

Life with AI: Creating human-AI systems that amplify intelligence, empathy, and human flourishing.

Each theme spans multiple domains and methods, fostering new forms of knowledge creation that are both academically robust and powerfully impactful in the real world.

Turning vision into practice

This vision comes to life through a distinctive academic model that blends deep research with creative inquiry and dynamic prototyping. The MIT Media Lab's academic foundation is the Program in Media Arts and Sciences (MAS), housed within the MIT School of Architecture and Planning (SA+P). The MAS program offers master's and doctoral degrees through a fully immersive, interdisciplinary, studio-based learning environment. Students are embedded in research groups from day one, contributing directly to innovative work that spans traditional boundaries.

MAS students are mentored by world-renowned faculty and supported by an ecosystem that encourages exploration, entrepreneurship, and experimentation. The result is a new kind of academic experience – highly collaborative, future-facing, and deeply humanistic.

Whether advancing new algorithms, engineering bionic limbs, composing immersive operas, or reimagining civic infrastructure, Media Lab students are co-authors of the research and creations that define their education.

A global community for transformative impact

The Media Lab is home to a diverse community of faculty, academic researchers, students, staff and alumni who are expanding the horizons of what's possible. From prototyping and deploying new technologies to publishing foundational research, this collective community drives the Lab's mission and shapes its culture of innovation – with students carrying that spirit forward into impactful careers in academia, industry, design studios, startups, and beyond.

Our collaborations with industry, NGOs, governments, foundations, cultural organizations, and philanthropists enable us to bring research out of the lab and into the world—driving change across sectors and geographies.

Research

Groups

affective computing

Advancing human wellbeing by developing new ways to communicate, understand, and respond to emotion

biomechatronics

Enhancing human physical capability

camera culture

Making the invisible visible–inside our bodies, around us, and beyond–for health, work, and connection

city science

Looking beyond smart cities

conformable decoders

Converting the patterns of nature and the human body into beneficial signals and energy

critical matter

Designing across the scale, imagining a transformative future

fluid interfaces

Designing systems for cognitive support

future sketches

Exploring the essence of code as a creative medium

human dynamics

Exploring how social networks can influence our lives in business, health, governance, and technology adoption and diffusion

lifelong kindergarten

Engaging people in creative learning experiences

molecular machines

Engineering at the limits of complexity with molecular-scale parts

Research (continued)

multisensory intelligence

Creating human-AI symbiosis across scales and sensory mediums to enhance productivity, creativity, and wellbeing

nano-cybernetic biotrek

Inventing disruptive technologies for nanoelectronic devices and creating new paradigms for life-machine symbiosis

opera of the future

Extending expression, learning, and health through innovations in musical composition, performance, and participation

personal robots

Building intelligent personified technologies that collaborate with people to help them learn, thrive, and flourish

responsive environments

Augmenting and mediating human experience, interaction, and perception with sensor networks

sculpting evolution

Cultivating wisdom through evolutionary and ecological engineering

signal kinetics

Inventing, building, and deploying wireless sensor technologies to address complex problems in society, industry, and ecology

space enabled

Advancing justice in Earth's complex systems using designs enabled by space

tangible media

Seamlessly coupling the worlds of bits and atoms by giving dynamic physical form to digital information and computation

viral communications

People and intelligent machines in a creative, symbiotic loop

Initiatives

Community Biotechnology Initiative Working at the intersection of community organizing and movement building, collective intelligence and social sciences, and accessible biotechnology tool development to activate the global movement around grassroots-driven life sciences, or “community bio”

Digital Currency Initiative Decentralizing trust and disrupting power structures with cryptographic peer-to-peer exchange and distributed systems

Space Exploration Initiative Building the technologies, tools, and human experiences of our sci-fi space future

MIT RAISE Initiative Advancing equity in learning, education, and computational action to rethink and innovate how to holistically and equitably prepare diverse K-12 students, an inclusive workforce, and lifelong learners to be successful, responsible, and engaged in an increasingly AI-powered society

Centers

MIT Center for Constructive Communication Designing tools, methods, and systems to understand and address societal fragmentation

K Lisa Yang Center for Bionics Pioneering transformational bionic interventions across a broad range of conditions affecting the body and mind

Programs

AHA Program Advancing Humans with AI

sAIpien Program Scalable AI Program for the Intelligent Evolution of Networks

WHx Program Unlocking women’s health through transformative technologies

Join us in inventing the future

If you're a prospective student seeking a rigorous yet creatively unconventional academic path, a business leader or policy-maker looking to uncover truly revolutionary ideas, or a supporter ready to invest in a better future – the MIT Media Lab welcomes you.

Together, we're not just imagining the future—we're building it.

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